Summary

Shift Cipher

- basically moves every plaintext letter by the same fixed amount.
- encryption → $\boxed{c = E(p) = (p+k) \pmod{26}}$
- decryption ⇒ $\boxed{p = D(c) = (c-k) \pmod{26}}$

  p: plain text, k: shif key, c: ciphertext value

- this can be easily broken by trying out all the 26 possible shifts
  - to basically bruteforce

Affine Cipher

- Here, we apply both addition & multiplication
- Encryption ⇒

  $\boxed{c = E(p) = (ap+b) \pmod{26}}$

  equivalent to linear equation: $y = ax + b$

  & to be reversible: $\gcd(a, 26) = 1$

  else, unique decryption is impossible

- Decryption ⇒

$$c \equiv ap + b \pmod{26}$$

$$c - b \equiv ap \pmod{26}$$

$$\boxed{p = D(c) = a^{-1}(c-b) \pmod{26}}$$

Chosen plaintext & chosen ciphertext attacks

- In a chosen plaintext attack, the attacker can select plaintext & observe the resulting ciphertext. Carefully selected plaintext letters can simplify the equations and allow the cipher to be recovered.

- in a chosen plaintext attack, the attacker selects ciphertext and observes the resulting plaintext.
the resulting equations can be used to recover parameters of the decryption function and then determine the encryption key

Vigener Cipher

- This works similarly to a shift cipher, but it uses a sequence of shifts rather than one fixed shift.

- A keyword is converted to numerical values : $k_1, k_2, \dots, k_m$

- if len(plaintext) > len(key) ⇒ repeat it from the beginning.

- Encryption ⇒ $\boxed{c_i = (p_i + k_i) \pmod{26}}$
- Decryption ⇒ $\boxed{p_i = (c_i - k_i) \pmod{26}}$

- However, if the key is short & repeated a lot, patterns are introduced. these might then be recovered by cryptanalysis.

XOR encryption ⇒

- encryption works bit by bit ⇒ $\boxed{C_i = P_i \oplus K_i}$
- it's equivalent to → $C_i = (P_i + K_i) \pmod 2$
- & same operation does both encryption & decryption ⇒ $C = P \oplus K$

$P = C \oplus K$

- this works becuse $K \oplus K = 0$ & $P \oplus 0 = P$.

Permutations

- just rearrange the position of the letter accordingly.
- A B C D

1 2 3 4 ← positions

for permutation key $(3, 1, 4, 2)$ ⇒ C A D B

Substitution Cipher

- replaces each plaintext symbol with another symbol
- mapping are very unique, not repeated at all, if so, will break
- follows a pre-defined set of substitutions/rules.
- basically like a 1-1 permutation of the alphabet.
- these are not secure & can be cracked with statistical analysis

<u>Key</u>

A → Z, B → P

C → M, D → 2

⇒

<u>example</u>

ABCD ⇒ ZPM2

by referring to the key

Block Cipher

- basically split the thing into fixed lengthed blocks & do the thing instead of dealing with individual letters
- block sizes are based on powers of two
- eg: hill cipher

Hill Cipher

- We use an invertible matrix as the encryption key
- for an $n \times n$ matrix, we divide the plaintext in n letters.
- each block is then (mod 26) to bring values to 0–25.

Encryption $\Rightarrow$ $\boxed{C = P \times M \pmod{26}}$

P: PT. row vector, M: matrix key, C: CT. row vector

- If column vectors are used, we change the multiplication order
- if PT is not divisible to blocks properly, add padding.

Example $\Rightarrow$ HI | RU , $M = \begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix}$

HI $\Rightarrow$ $[7 \quad 8]$ RU $\Rightarrow$ $[17 \quad 20]$

$$[7 \quad 8]\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} = [13 \quad 18] \Rightarrow \text{NS}$$

$$[17 \quad 20]\begin{bmatrix} 1 & 2 \\ 4 & 7 \end{bmatrix} = [19 \quad 18] \Rightarrow \text{TS}$$

$\therefore$ HIRU $\Rightarrow$ NSTS

- Decryption $\Rightarrow$

$$\boxed{P = C \times M^{-1} \pmod{26}}$$

- only possible when $\gcd(\det(M), 26) = 1$.
- if gcd is not 1, that matrix X be used for unique decryption

- inverse of a 2x2 matrix $\Rightarrow$

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad , \quad \det(M) = ad - bc$$

and $\boxed{M^{-1} = (\det M)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}}$

- inverse of any larger matrix is

$$\boxed{M^{-1} = (\det M)^{-1} \operatorname{adj}(M) \pmod{26}}$$

where $\operatorname{adj}(M) = [C(M)]^T$

- Recovery an unknown Matrix $\Rightarrow$

- need to know both PT (matrix A) and CT (matrix B)

$A \times M = B \pmod{26}$

if A is invertible $\quad A^{-1} \times A M = A^{-1} B$

therefore $\quad \boxed{M = A^{-1} B \pmod{26}}$

also, PT must satisfy $\quad \gcd(\det A, 26) = 1$

otherwise, A^{-1} does not exist

- Example $\Rightarrow$

S I T	C R Y	P T O
C I H	O L I	N F O

3x3 matrix

SIT $\Rightarrow$ [18 8 19]
CRY $\Rightarrow$ [2 17 24]
PTO $\Rightarrow$ [15 19 14]

$$\therefore A = \begin{bmatrix} 18 & 8 & 19 \\ 2 & 17 & 24 \\ 15 & 19 & 14 \end{bmatrix}$$

CIH $\Rightarrow$ [2 8 7]
OLI $\Rightarrow$ [14 11 8]
NFO $\Rightarrow$ [13 5 14]

$$\therefore B = \begin{bmatrix} 2 & 8 & 7 \\ 14 & 11 & 8 \\ 13 & 5 & 14 \end{bmatrix}$$

$$A \times M = B$$

$\times A^{-1}$ $$\underbrace{A^{-1} A}_{} M = A^{-1} B$$

$I \Rightarrow$ the identity matrix

$$M = A^{-1} B$$

find $C(A) \Rightarrow C(A) = \begin{bmatrix} 16 & 20 & 17 \\ 15 & 19 & 12 \\ 25 & 22 & 4 \end{bmatrix}$, by solving $C_{11}, C_{12}, C_{13}, C_{21}, \dots, C_{33}$

find $Adj(A) = \begin{bmatrix} 16 & 15 & 25 \\ 20 & 19 & 22 \\ 17 & 12 & 4 \end{bmatrix}$

find $\det(A) = 18C_{11} + 8C_{12} + 19C_{13} = 771 \bmod 26 = 17$

find 17^{-1} with extended eucledian algorithm

$\Rightarrow 1 = 2 \times 26 - 3 \times 17$

$\therefore -3 \times 17 = 1 \pmod{26} \Rightarrow 17^{-1} \equiv -3 \pmod{26}$

$\equiv 23 \pmod{26}$

$$\therefore \quad A^{-1} = 23 \times \text{Adj}(A) \pmod{26}$$

$$A^{-1} = \begin{bmatrix} 4 & 7 & 3 \\ 18 & 21 & 12 \\ 1 & 16 & 14 \end{bmatrix} \pmod{26}$$

and $$M = A^{-1} \times B = \begin{bmatrix} 15 & 20 & 22 \\ 18 & 19 & 20 \\ 18 & 20 & 19 \end{bmatrix} \pmod{26}$$

regular matrix multiplication

Learning Reflection

The most important thing I learned from this task was how different classical cipher systems work. Specially affine and hill ciphoers, and how modular arithmetic is used to encrypt and decrypt messages.

This relates to what I already know about basic encryption and discrete mathematics, but this module showed me how concepts like matrices, modular inverses and the eucledian algorithm actually apply in real world cryptographical concepts.

I also really enjoyed the cryptanalysis parts.

I think this is important for my degree because it gives a better understanding how cryptography works underneath the hood, instead of just using encryption tools without understanding the actual maths behind them.